9.11 Collecting the Set

WORKSHEET

The question

STEP II 1994, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/94-s2-q13

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps. There is space after each hint for your working.

Ask yourself If each packet independently gives a new card with probability p, what is the expected number of packets you open up to and including the first new card?

STAGE 1 Waiting for a new card

Covers: waiting for a new card

Ask yourself With r different cards already, what is the chance that the next packet gives a new one? Use the expected wait for a geometric distribution.

STAGE 2 Comparing a sum with an integral

Covers: the integral inequality

Ask yourself On $r \leq x \leq r+1$, between which two values does $\frac{1}{x}$ lie? Integrate over this interval of length 1.

Pause and try Add your inequalities from $r = 1$ to $r = n-1$. The integrals join up into one integral: which one?

STAGE 3 Completing the set

Covers: collecting the whole set

Ask yourself The total number of packets is the sum of the waits for each new card. Add their expectations, and use the second stage to compare the sum with $\ln n$.

What's really going on

How good is the estimate?

The comparison can be sharpened. The shaded gaps between the curve and the upper rectangles in Figure 9.11.1 (in the full section online) have a finite total area, so $H_n - \ln n$ tends to a limit,

$$\gamma = \lim_{n\to\infty}(H_n - \ln n) = 0.5772156649\ldots,$$

Euler's constant. (Nobody knows whether γ is rational.) So the expected number of packets is $n\ln n + \gamma n + \frac{1}{2} + \cdots$. For the 600-sticker album, $600 \times (\ln 600 + 0.577) \approx 4185$, which is the figure quoted in the hook.

Where does the time go? The stages are very unequal. Collecting the first half of the set takes only about $n\ln 2 \approx 0.69n$ packets, while the last card alone takes n on average. Most of the effort goes into the last few cards, which is why collectors swap duplicates.

Concentration

The expected value is only the start: how much does the total vary? The variance of the total is $\sum nr/(n-r)^2$, which is about $\frac{\pi^2}{6}n^2$ (the sum $\sum 1/k^2 = \pi^2/6$ from Section 2.4 appears). So the standard deviation is about $1.28n$, much smaller than the mean $n\ln n$ when n is large: the total is *concentrated* around its mean. Paul Erdős and Alfréd Rényi showed in 1961 that, more precisely, the total minus $n\ln n$, divided by n, has a limiting distribution, the *Gumbel distribution*, which also describes the largest of many random quantities (Section 12.25).

Coupons everywhere

The problem goes back to Abraham de Moivre (1712) and Laplace. It now appears in many disguises. How many random tests are needed to check every component of a system? How long until a random walk on a network has visited every node? How many times must a computer hash table be probed? In each case the answer involves $n \ln n$. In biology, the same calculation estimates how many sequencing "reads" are needed to cover a whole genome.

Try this next

1. **Half a set.** Show that the expected number of packets to collect $\frac{n}{2}$ different cards (for even n) is $n(H_n - H_{n/2})$, and that this is about $n \ln 2$.

2. **Distinct cards.** After opening k packets, show that the expected number of different cards you have is $n(1 - (1 - \frac{1}{n})^k)$. (*Hint:* for each card, find the probability that you have it, and add.) Roughly how many packets give you 95% of the set?

3. **Two collectors.** Two children collect the same set, sharing packets so that a duplicate for one may be new for the other. Show that the expected number of packets to complete both sets is about $n(\ln n + \ln \ln n)$. (This is harder; first convince yourself that it is much less than $2n \ln n$.)

4. **A sharper bound.** Use the trapezium rule picture instead of rectangles to show that $H_n - \ln n$ is decreasing, and that $H_n - \ln n - \frac{1}{2n}$ is increasing. Deduce $0.5 < \gamma < 0.7$ from small values of n.

Further reading

- **Geoffrey Grimmett & David Stirzaker, *One Thousand Exercises in Probability*** (Oxford University Press). Many variations on the coupon collector.
- **Julian Havil, *Gamma: Exploring Euler's Constant*** (Princeton University Press). Everything about γ and the harmonic series, at an accessible level.
- **Michael Mitzenmacher & Eli Upfal, *Probability and Computing*** (Cambridge University Press). The coupon collector and its uses in computer science.

The full guided solution is in Section 9.11:
https://beyondstep.keithmcnulty.org/ch09/coupon-collector.html